Software for a better world

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CS5741 Concurrency & Parallelism in Software Development

Lecture 1: Overview

Dr Faeq Alrimawi

HOST INSTITUTION

UNIVERSITY OF LIMERICK OLLSCOIL LUIMNIGH

PARTNER INSTITUTIONS

NUI Galway OÉ Gaillimh · Maynooth University · Trinity College Dublin, Coláiste na Tríonóide, Baile Átha Cliath, The University of Dublin · University College Dublin, An Coláiste Ollscoile, Baile Átha Cliath · UCC · MTU, Ollscoil Teicneolaíochta na Mumhan, Munster Technological University · DCU · Waterford Institute of Technology · Dundalk Institute of Technology, Dún Dealgan · LIT · GMIT

The Team

Module Leader: Faeq Alrimawi

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- PhD in ML/AI, UL
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Module Overview

- Aim & Outcomes
- Content
- Lectures & Labs
- Assessment
- Resources
- Brightspace Participation

Aim

To **introduce fundamental concepts/theories** of parallel and concurrent software development

Discuss how **designs** can be made parallel and identifies the **common faults** in concurrent programs and **how to avoid them**.

Introduce a range of widely used programming **techniques** for writing concurrent and large scale parallel systems.

Outcomes

- **Demonstrate** how the **key concepts** of parallelism and concurrency are of use when designing and developing software solutions.
- **Identify** and exploit **opportunities** for concurrency and parallelisation when developing a software system.
- **Avoid** or reduce the likelihood of **common problems** associated with parallel and concurrent systems.

Content

- **Key terminology**, Concurrency and parallelism, and applications to software
- **Methods** to identify parallelisable problems
- **Parallel algorithms** and how to design them
- **Common faults** in concurrent and parallel programs
- **Scheduling techniques**, complexity and performance measurement
- **Multi-core architectures** and programming paradigms
- **Data-parallel techniques**: SIMD, MIMD, vector processing and GPGPUs
- **Shared-memory techniques**: threads and locks, structured parallelism, OpenMP
- **Message-passing techniques** using MPI (Message Passing Interface)

Lecture & Lab

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	11:00 - 13:00 CS5741 - LAB - 2A FAEQ ALRIMAWI CS2044 Wks:2-12			13:00 - 15:00 CS5741 - LAB - 2B FAEQ ALRIMAWI CS2044 Wks:2-12	
			15:00 - 17:00 CS5741 - LEC FAEQ ALRIMAWI S207 Wks:1-12		

No Labs This Week

Assessment

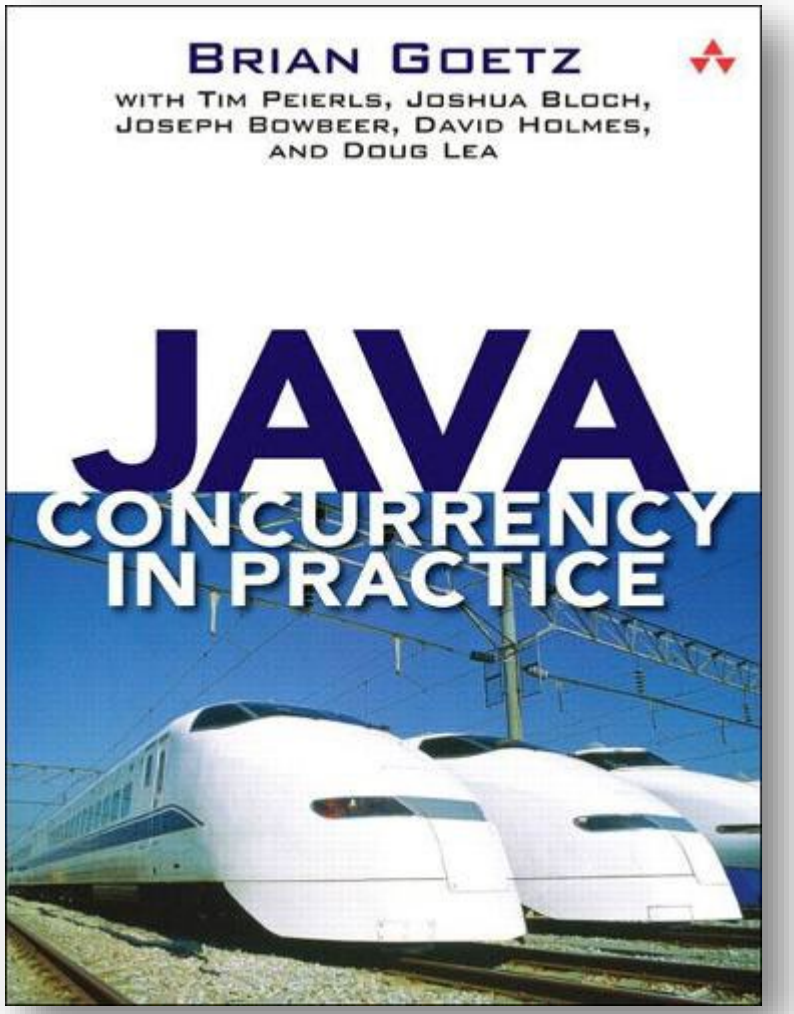
Type	Due	Percentage (100%)
Assignment – 1	Week 6-7	30%
Assignment – 2	Week 11-12	30%
Final Written Exam	Week 14/15	40%

Assignments are **Individual**

Resources

- Java Concurrency in Practice
  by Brian Goetz
  - Originally published in 2006, but still relevant
  - Can be found online

- Online Resources
  - Parallel Computing by Stewart Weiss: https://www.cs.hunter.cuny.edu/~sweiss/course_materials/csci493.65/csci493.65_lecture_notes.php



Other Resources

- **Module Material**: available on Brightspace
  - Lecture slides
  - Code segments

- **Web resources** referred to throughout the module
  - The material in these resources will be **examinable** also

Grading Scheme

Mark	Grade
80+	A1
72+	A2
64+	B1
60+	B2
56+	B3
52+	C1
48+	C2
40+	C3
35+	D1
0	F

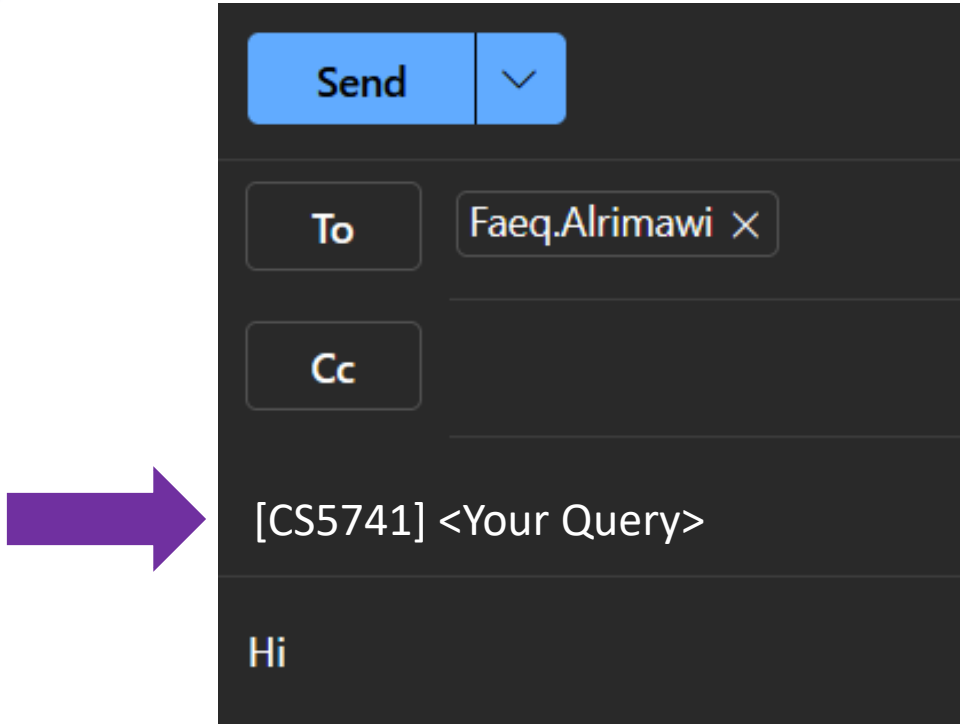
Brightspace Participation

Brightspace is an important part of your learning journey

- Help create a **collaborative community** where you can ask and answer questions
- Give you the opportunity to clarify general points **with your module leader/TA** in an open environment
- By asking these questions on Brightspace, rather than in an email, you **allow your peers to also benefit** from the answers

If you email me…

Please use subject line
[CS5741]



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What Are Your Questions?

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